CHRISTOPHER LOORKHOOR

Academic Curriculum Vitae • Chemical Engineering

Houston, TX | 832-870-7134 | cloorkhoor@outlook.com | linkedin.com/in/cloorkhoor

EDUCATION

University of Houston | Cullen College of Engineering | Expected May 2028

Bachelor of Science in Chemical Engineering • Houston, TX

RESEARCH INTERESTS

Heterogeneous catalysis • Reaction engineering • Catalyst synthesis and characterization • Sustainable fuels and chemical processes

RESEARCH EXPERIENCE

Undergraduate Researcher | CO_2 Adsorption & Sorbent Materials | Aug 2025–Present

Bollini Research Group, University of Houston | Advisor: Dr. Praveen P. Bollini

- Developing a comparative experimental framework to evaluate CO_2 adsorption on Ni–Ce hydroxides and calcined oxides at 25°C and 50°C, informed by published sorbent studies.
- Planning powder XRD and N_2 physisorption (BET) characterization to relate crystalline phases and textural properties to adsorption performance.
- Performed fixed-bed breakthrough tests on benchmark sorbents with CO_2 adsorption at 300°C and regeneration at 500°C; employed controlled gas feeds and online mass spectrometry to quantify uptake.
- Refined sorbent pretreatment, regeneration, and breakthrough-integration procedures to improve consistency in experimental analysis.

Undergraduate Researcher | Ga-Modified ZSM-5 Catalysis | Jun–Aug 2025

Bollini Research Group, University of Houston | Research mentor: Varad Joshi

- Investigated gallium incorporation in ZSM-5 for aromatic hydrocarbon formation from renewable feedstocks; prepared catalysts by liquid-phase ion exchange, drying, and calcination.
- Conducted continuous-flow fixed-bed ethylene aromatization experiments using online GC and manual chromatogram peak integration to quantify conversion and product selectivity.
- Characterized catalysts with powder XRD and N_2 physisorption (BET), identifying features consistent with Ga_2O_3 formation, surface area around 291 m^2/g, and a Type IV adsorption isotherm.

Computational Researcher | UH iGEM | Jun–Aug 2026

University of Houston | Metabolic modeling

- Developed Python/COBRApy metabolic models comparing citrate synthase variants across 108 computational scenarios to evaluate changes in predicted TCA-cycle flux.
- Predicted a 74% average increase in α-ketoglutarate production for R163L relative to native GltA under modeled conditions.

RESEARCH PRESENTATIONS

Poster — Loorkhoor, C., and Bollini, P. “Incorporation of Gallium on H-ZSM-5 for Jet Fuel Production.” Undergraduate Research Day, University of Houston, Houston, TX, April 9, 2026. Research mentor: Varad Joshi.

Oral presentation — Loorkhoor, C. “Incorporation of Ga in Jet Fuel Production on ZSM-5.” Internal Bollini Research Group meeting, Department of Chemical Engineering, University of Houston. (Date not specified.)

HONORS, AWARDS & FELLOWSHIPS

Stutts Research Program Scholar — University of Houston (2025) • Summer Undergraduate Research Fellowship (SURF) — University of Houston (2025) • Dean’s List — Fall 2024

TECHNICAL SKILLS

Laboratory & characterization: Fixed-bed reactors; gas-flow control; catalytic testing; CO_2 breakthrough analysis; online GC; online MS; powder XRD; N_2 physisorption/BET.

Modeling & computation: Python; MATLAB; COBRApy; Aspen Plus; Excel; numerical integration; regression; curve fitting; data visualization.

Research methods: Experimental design; literature analysis; catalyst selectivity and conversion analysis; research posters and oral presentations.

TEACHING EXPERIENCE

Teaching Assistant | Human Situation, Honors College | Aug 2025–Present

University of Houston

- Supported student communication, attendance management, course coordination, and examination logistics.
- Developed a Python-assisted scheduling workflow serving more than 300 students and 10 faculty members.

SELECTED ENGINEERING PROJECT

Steam Methane Reforming Process Simulation | Aspen Plus | Sep–Dec 2025

- Built an Aspen Plus flowsheet incorporating reactors and separation operations to model steam methane reforming and water-gas-shift reactions for hydrogen production.
- Evaluated a conceptual five-unit process targeting approximately 1.63×10^5 kg/day hydrogen and documented process assumptions in a technical memo.

PROFESSIONAL DEVELOPMENT

AIChE/CCPS SAChE — Six process-safety certificates (2026), covering hazard recognition, hazard identification and risk assessment (HIRA), and PHA/HAZOP/LOPA concepts.